

Source-Seeded Weak-Mode Support Laws

Coarse-Boundary Quotients and a Finite-Extrapolation Barrier

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Abstract

RH-106 identified the uniform quotient price as c^2/g , where c is an omitted cross energy and g is a retained-to-omitted spectral gap. The next question is whether this price must be controlled at every small-noise level. For the source-seeded adaptive rule with maximum enrichment width four, we show that the answer can instead be reduced to a support question.

Let $s_1 \geq s_2 \geq s_3 \geq s_4$ be the four projected-cross singular values and let τ be the relative cutoff. The exact selector identity is

$$w_\tau < 4 \iff \frac{s_4}{s_1} < \tau.$$

Consequently, if the fourth-cross ratio stays above the cutoff beyond a finite level k_* , the quotient support is empty there and the accumulated quotient price reduces to a finite coarse-layer sum. We prove the resulting support-to-price reduction and combine it with the stopped endpoint theorem: if the finite coarse price fits the endpoint allowance, no later quotient price can accumulate; if it does not, the stopped fallback remains safe.

The five source-seeded scales exhibit a sharp coarse boundary. At cutoffs $10^{-8}, 10^{-6}, 10^{-4}$, the weak-mode events are respectively

$$(4, 1, 0, 0, 0), \quad (7, 2, 2, 0, 0), \quad (8, 8, 6, 0, 0)$$

across $\sigma = (0.16, 0.08, 0.04, 0.02, 0.01)$. Thus even the 10^{-4} cutoff has no quotient event at the last two scales. The minimum fine-scale support margin is 7.3649 times the cutoff. All 360 selector comparisons agree with the recorded adaptive widths, all 38 local gap certificates are green, and stopped endpoint ratios remain below 1.006033.

This is a positive sparse-supply result, not an all-level asymptotic theorem. We prove a finite-extrapolation barrier: separated and persistent-support cross-ratio sequences can agree on every finite audit and diverge afterward. Therefore the observed coarse boundary must ultimately be supplied by an analytic support-separation theorem. No Stage A, Hilbert–Polya, zero identification, or Riemann Hypothesis result is claimed.

Keywords: source-seeded packet; weak-mode support; adaptive Ritz width; quotient price; sparse supply; finite extrapolation.

MSC 2020: 47A75; 15A18; 65F15; 65G20; 37C30.

1 Introduction

The source-seeded refresh construction tracks a finite packet from the source coordinates rather than repeatedly seeding late ambient directions [1]. RH-96 used that packet to test adaptive weak-mode quotients. Its local theorem was favorable: an omitted block with coupling C and retained-to-omitted gap g costs at most $\|C\|_F^2/g$ [3]. Its global experiment was more subtle. A relative cutoff of 10^{-8} was safe, while more aggressive cutoffs failed after recursive accumulation even though every local certificate remained green.

RH-102 resolved endpoint safety by pricing the actual nonlinear future and stopping before the absolute debit exhausts the endpoint slack [2]. RH-106 then wrote an abstract no-stop condition in terms of candidate count, propagation factor, local price, and slack. It left open whether the candidate count itself grows through the fine scales.

This paper isolates a different route to uniformity. The adaptive rule has a hard maximum width. If the fourth projected-cross mode is not weak, no quotient is proposed at all. Hence a lower bound on the fourth-cross ratio, not a lower bound on the retained-to-omitted Ritz gap, can eliminate the fine part of the quotient problem.

The contributions are:

1. an exact adaptive-support equivalence for the four-direction selector;
2. a support-to-price reduction theorem and its stopped endpoint corollary;
3. a finite-extrapolation barrier showing why five anchors cannot prove eventual support separation;
4. a complete audit of the 120-update source-seeded chains at three thresholds.

2 Adaptive support identity

At one source-seeded update, let the projected-cross singular values be

$$s_1 \geq s_2 \geq s_3 \geq s_4 \geq 0, \quad s_1 > 0. \quad (1)$$

For a relative cutoff $\tau > 0$, the maximum-width-four selector used in RH-96 is

$$w_\tau = \max\{2, \min(4, \#\{j \in \{1, 2, 3, 4\} : s_j/s_1 \geq \tau\})\}. \quad (2)$$

Definition 2.1 (Weak-mode support). The weak-mode support at cutoff τ is

$$\mathcal{W}_\tau = \left\{ t : \frac{s_4(t)}{s_1(t)} < \tau \right\}.$$

The support margin at a nonweak update is

$$m_\tau(t) = \frac{s_4(t)/s_1(t)}{\tau}.$$

Theorem 2.2 (Adaptive support equivalence). *Under (1) and (2),*

$$w_\tau < 4 \iff t \in \mathcal{W}_\tau \iff \frac{s_4(t)}{s_1(t)} < \tau. \quad (3)$$

In particular, if $s_4(t)/s_1(t) \geq \tau$ on a collection of updates, all those updates use the full width four and generate no omitted-mode quotient.

Proof. Since the singular values are ordered, the number of indices satisfying $s_j/s_1 \geq \tau$ is four exactly when the fourth ratio is at least τ . The outer maximum and minimum in (2) do not change the equivalence between width four and failure of that condition. $\square$

The theorem is elementary but important for the route. It converts an adaptive spectral statement into a binary support statement before any gap certificate is invoked.

3 Support-to-price reduction

Index the source-seeded noise levels by k and updates within one level by t . At a weak update, let $\ell_{k,t} = c_{k,t}^2/g_{k,t}$ be the local gap-weighted price and let $K_{k,t}$ be a certified propagated-debit factor, so that the endpoint debit satisfies

$$d_{k,t} \leq K_{k,t}\ell_{k,t}. \quad (4)$$

Define the total quotient price

$$D_k(\tau) = \sum_{t \in \mathcal{W}_{k,\tau}} K_{k,t}\ell_{k,t}. \quad (5)$$

Theorem 3.1 (Coarse-support-to-price reduction). *Suppose there is a level k_* such that*

$$\frac{s_4(k,t)}{s_1(k,t)} \geq \tau \quad \text{for every } k \geq k_* \text{ and every update } t. \quad (6)$$

Then

$$D_k(\tau) = 0 \quad (k \geq k_*). \quad (7)$$

If the finitely many coarse levels $k < k_$ have at most N_* weak events, $K_{k,t} \leq K_*$, and $\ell_{k,t} \leq \ell_*$, then*

$$\sup_{k \geq 0} D_k(\tau) \leq N_* K_* \ell_*. \quad (8)$$

Proof. Theorem 2.2 makes the support set empty for every $k \geq k_*$. This proves (7). On the remaining finite set of levels, the sum in (5) contains at most N_* terms, each bounded by $K_*\ell_*$. Taking the supremum proves (8). $\square$

Corollary 3.2 (Support-reduced stopped endpoint law). *Let S_k be the rigorous endpoint slack and $A_k = \eta S_k$ the stopped allowance, with $0 \leq \eta < 1$. Under the hypotheses of Theorem 3.1, if*

$$N_* K_* \ell_* \leq \inf_{k < k_*} A_k,$$

every coarse quotient and hence every candidate in the family is accepted without a stop, and every endpoint gate is preserved. If the inequality fails, accepting only while the cumulative propagated debits fit A_k and switching permanently to full width still preserves every endpoint gate.

Proof. The first statement follows by summing the finite coarse prices and applying the endpoint-budget theorem of RH-102. The second is its stopped branch; the fine levels contribute zero quotient price by (7). $\square$

Remark 3.3. This reduction is different from proving a fine-scale lower bound on the retained-to-omitted Ritz gap. If no fourth mode is omitted, that gap never enters a quotient price. The physical target has therefore split into a cross-support separation problem and a finite coarse price problem.

4 Finite-extrapolation barrier

The support reduction is conditional. The next proposition states exactly why the five finite anchors cannot, by themselves, establish (6) for all future levels.

Proposition 4.1 (Finite support-extrapolation barrier). *Fix a cutoff $\tau > 0$ and any finite observed sequence $(\theta_0, \dots, \theta_K)$ with $\theta_k > 0$. There are two infinite extensions agreeing with all observed values:*

$$\theta_k^{\text{sep}} = 2\tau, \quad k > K, \quad (9)$$

$$\theta_k^{\text{pers}} = \frac{1}{2}\tau, \quad k > K. \quad (10)$$

The first has empty weak support beyond K , while the second has a weak event at every later level. Thus no finite cross-ratio audit alone proves eventual support separation.

Proof. Define the two extensions as displayed. Since $2\tau \geq \tau$ and $\tau/2 < \tau$, the adaptive support equivalence gives the two claims. $\square$

Remark 4.2. The barrier is not a negative result about the source-seeded family. It identifies the exact missing analytic input: a structural lower bound for the fourth projected-cross ratio on the fine source-seeded chain.

5 Five-scale source-seeded audit

We read all 120 updates in both channels at the five RH-94 source-seeded scales, for each of the three RH-96 cutoffs. The audit recomputes s_4/s_1 , applies the selector identity, and independently reads the local coupling/gap price and RH-102 propagated debit. There are 360 selector comparisons in total.

Table 1: Weak-mode support and coarse price by relative cutoff. Event counts are listed in scale order 0.16, 0.08, 0.04, 0.02, 0.01.

τ	event counts	fine level	min fine margin	total ℓ	max stopped ratio
10^{-8}	(4, 1, 0, 0, 0)	$k \geq 2$	8.221	$4.297 \cdot 10^{-13}$	1.001173
10^{-6}	(7, 2, 2, 0, 0)	$k \geq 3$	736.493	$1.452 \cdot 10^{-9}$	1.001173
10^{-4}	(8, 8, 6, 0, 0)	$k \geq 3$	7.365	$7.500 \cdot 10^{-8}$	1.006033

Every selector comparison agrees with the recorded adaptive width. The fine support is empty for every threshold. The absolute fourth-ratio lower bound on the fine levels is 8.221×10^{-8} for 10^{-8} , and 7.365×10^{-4} for the two coarser cutoffs. Thus the weakest observed fine support margin is still 7.365.

The local prices are concentrated in the coarse layer. The maximum total local price over the three thresholds is 7.5000×10^{-8} . Using the largest finite replay multiplier, the coarse support upper is 6.702×10^{-8} . The actual stopped chains are much tighter: all 38 local gap certificates are green, 35 candidates are accepted, three are rejected, and the worst stopped endpoint/reference ratio is 1.006033.

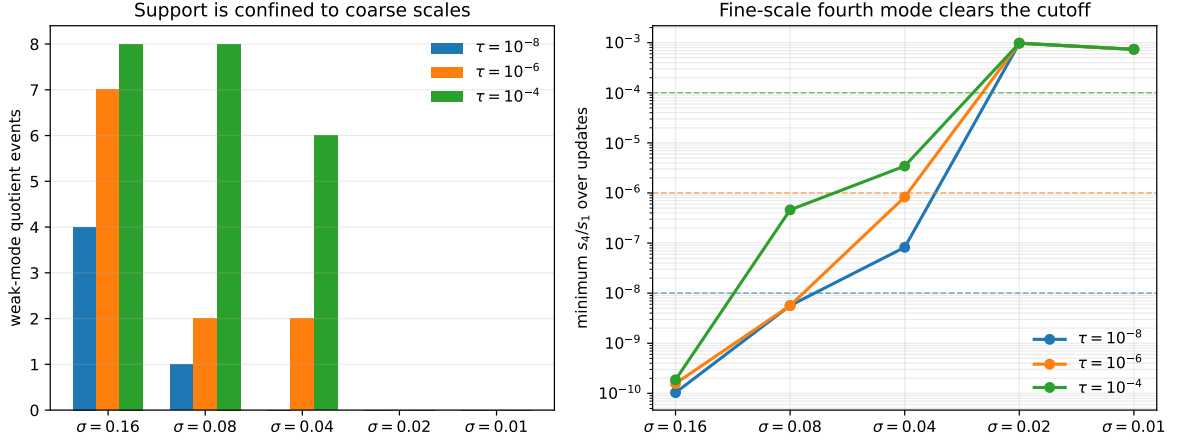


Figure 1: Left: weak-mode quotient events occur only on the coarse scales. Right: the minimum fourth-cross ratio over each scale’s updates clears the corresponding cutoff on the fine scales.

The finite data therefore support a useful route refinement: at the current cutoffs, fine-scale quotient accumulation is absent, not merely small.

6 Route consequence and claim boundary

RH-107 supplies a new conditional decomposition of the uniform quotient gate:

1. prove a fine source-seeded support-separation bound for s_4/s_1 ;
2. price only the finite coarse weak-mode layer;
3. use the stopped clock as a safety fallback if the coarse price does not fit.

This is more precise than asking for a uniform gap lower bound at every level. It also explains why the finite RH-96/RH-102 chains look easier at the fine scales: the quotient branch is not being entered there.

The analytic support-separation statement remains open because of Proposition 4.1. Until it is proved, the result is a finite sparse-supply certificate and a route map, not an all-level theorem. The remaining absolute-scale leaves from RH-104 and RH-105 are unchanged.

No statement here constructs a Hilbert–Polya operator, proves a $T \log T$ law or prime-power trace formula, identifies zeta zeros, or proves the Riemann Hypothesis.

7 Reproducibility

The directory contains the support selector helpers, full and smoke audits, figures, tests, and a hash-checked archive. Run:

```
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \\  
  experiments/build_support_audit.py  
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \\  
  experiments/build_support_audit.py --smoke  
MPLBACKEND=Agg PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \\  
  experiments/make_figures.py  
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/pytest -q -p no:cacheprovider
```

The archive scripts record the exact RH-96/RH-102 inputs and verify all publication hashes.

References

- [1] Liang Wang. Source-seeded four-direction horizon refresh. Preprint and software repository, RH-94, 2026.
- [2] Liang Wang. Stopped hybrid quotient clocks. Preprint and software repository, RH-102, 2026.
- [3] Liang Wang. Gap-weighted weak-mode quotients. Preprint and software repository, RH-96, 2026.